BUSINESS REQUIREMENTS DOCUMENT

Intern Project Management and Performance Tracking Platform

Cross-Checked Revision — Version 3.0

Prepared against BRD v2.1, implementation audit, HLD and LLD

Date: 9 October 2026 | Status: Draft for mentor review

Document type	Business Requirements Document
Version	3.0 — Cross-Checked Draft
Baseline	BRD_PRMS_v2.1, implementation audit, HLD and LLD
Purpose	Define business scope, functional requirements, business rules, access and reporting expectations
Status	Draft for mentor review; open decisions are listed in Section 12

1. Scope

1.1 Problem Statement

Internship and project work can be distributed across conversations, meetings, forms, and separate task-tracking channels. This makes ownership, deadlines, submissions, time spent, and progress harder to trace. Managers and technical leads need a consolidated, permission-controlled view of project execution, intern work, and outstanding commitments.

1.2 Proposed Solution

Provide a centralized, organization-scoped platform for user administration, project planning, hierarchical work management, sprint execution, task-level time tracking, forms and submissions, notifications, cohort management, and role-appropriate dashboards. Manager and intern experiences shall use the same authoritative application data and compatible backend services.

1.3 In-Scope Capabilities

- Authentication, profile management, organization-scoped access, and permission-controlled actions.
- Organization and user administration, including member provisioning, batch/sub-batch and cohort grouping, and authorized bulk import.
- Project and project-member management, including project-level responsibilities.
- Hierarchical work items with assignments, priority, status, dates, progress, story points, and bug tracking.
- Project Roadmap, Backlog/Epics, Sprint Workspace, and Kanban views that remain synchronized over shared work-item data.
- Sprint creation, activation, completion, and handling of unfinished work.
- Task start/stop work sessions, manual time entries where permitted, and timesheet reporting.
- Form creation, publication, targeted assignment, submission, response review, and submission-status visibility.
- In-app notifications and broadcast announcements within authorized scope.
- Action Items for follow-up commitments that must be completed before the next meeting.
- Manager and intern dashboards, project progress visibility, deadlines, and cohort-scoped views.
- Work-session-derived time and attendance visibility, subject to the attendance policy decisions in Section 12.

1.4 Not Included as Confirmed Current Functionality

- Minutes of Meeting (MoM) authoring or meeting-minutes management. Action Items are a separate follow-up workflow and do not mean meeting minutes.
- Fully implemented advanced performance scoring and performance-reporting analytics.
- Automated scheduled deadline notifications unless separately implemented and verified.
- Dedicated manual attendance check-in/check-out workflows, biometric attendance, or IP-based attendance.
- External integrations not verified in the implementation audit.

2. Business Objectives

- Centralize project and internship work, ownership, deadlines, forms, notifications, and follow-up commitments.
- Maintain a traceable hierarchy for project work and enforce valid parent-child relationships.
- Give each user visibility into work appropriate to their role, organization, project membership, and assignments.
- Allow teams to plan and execute sprints while keeping Roadmap, Backlog, Sprint Workspace, and Kanban data synchronized.
- Record actual time spent on tasks and produce transparent timesheet summaries.
- Make pending, overdue, blocked, completed, and unsubmitted work visible to authorized users.
- Provide reliable cohort and member management without duplicating users or project data.

3. Work Management Structure

Project work shall use the following hierarchy: Project → Epic → Story or Bug → Task → Subtask, subject to the specific parent-child rules below. Manager personal tasks are separate from project hierarchies.

Level / Type	Business rule
Project	Top-level container for project work; project root work items are Epics.
Epic	May contain Stories or Bugs.
Story	Must belong to an Epic; may contain Tasks or Bugs.

Level / Type	Business rule
Task	Must belong to a Story; may contain Subtasks.
Subtask	Must belong to a Task and cannot have children.
Bug	May belong to an Epic or Story and cannot have children.
Manager personal task	A standalone Task without a project or parent.

3.1 Hierarchy Integrity

- Parent and child items must belong to the same project and organization.
- The system shall reject invalid parent-child combinations and circular parent relationships.
- Changing an item's type must not invalidate its parent or existing children.
- Deletion behavior must preserve hierarchy integrity and be confirmed before destructive descendant removal. The LLD's descendant-deletion test is marked Planned, so this behavior is not treated as verified.

3.2 Work Item Attributes

Work items shall support, where applicable, a title, description, type, status, priority, assignee(s), start date, due date, completion timestamp, sprint, story points, and parent-child relationships. The system shall identify overdue work when its due date has passed and it is not completed.

4. Sprints and Project Workspaces

4.1 Project Views

- Roadmap: timeline visibility for Epics and their direct children.
- Backlog and Epics: hierarchical work-item planning, filtering, estimation, and sprint allocation.
- Sprint Workspace: execution view for the active sprint, with status columns and logged-work visibility.
- Kanban Board: continuous-flow work board with status changes and filters such as My Tasks, Overdue, and Blocked.
- Action Items: follow-up commitments due before the next meeting; see Section 7.

4.2 Sprint Lifecycle

- Authorized users shall create sprints with a name, goal, and planned date range.
- A project shall have no more than one active sprint at a time.
- Users shall be able to start a planned sprint and complete an active sprint.
- When completing a sprint, authorized users shall choose whether unfinished items return to the backlog or move to another eligible sprint in the same project.
- Story points and completed work shall be visible for sprint planning and review.

4.3 Burndown and Progress

The sprint workspace shall show actual remaining work against an ideal pace over the sprint date range. Remaining work shall use story points where estimates are available, or item count where estimates are unavailable. Progress and sprint-health metrics must be calculated from consistent sprint-scoped data. Historical progress must not be fabricated when historical status events or snapshots are unavailable.

5. User Roles, Organizations, and Access

The platform shall separate organization/system roles from project-membership roles. Permissions and project/work scope shall determine which actions a user can perform. The exact mapping of business titles to implemented roles requires mentor confirmation.

Access layer	Roles evidenced by implementation audit	Notes
System / organization	SUPER_ADMIN, ADMIN, MANAGER, TECH_LEAD, EMPLOYEE, INTERN	Audit reports these as implemented system roles; exact permission mapping must be maintained from backend authorization.
Project membership	DEVELOPER, QA_TESTER, TECH_LEAD, UI_UX_DESIGNER, DEVOPS_ENGINEER	Project-level membership roles; not interchangeable with system roles.
Earlier BRD business titles	Technical Architect, Capability Lead, Mentor	Not found as implemented system roles in the audit. Confirm whether these map to existing roles or should be introduced.

5.1 Access Requirements

- Protected functionality shall require authentication.
- Users shall only access organizations, projects, work items, forms, documents, and reports within their authorized scope.
- Permissions shall be enforced by backend services, not only by hiding frontend controls.
- Managers shall have authorized organization-level oversight and user/project administration.
- Tech Leads shall have only the project and task permissions granted by the system.
- Interns and Employees shall access permitted project information and their authorized work, submissions, notifications, and time-tracking features.
- Super Admin and Admin capabilities shall be constrained to their respective administrative responsibilities.

5.2 Organization and Cohort Management

The platform shall support organization-scoped records and cohort/batch/sub-batch grouping. Authorized managers shall be able to search members and filter relevant dashboards and member rosters by cohort. Cohort filtering must not expose data outside the user's organization or permissions.

6. Functional Requirements

6.1 User and Organization Management

ID	Requirement
FR-UM-01	The system shall authenticate users using unique accounts and protect authenticated routes and APIs.
FR-UM-02	The system shall allow users to view and update permitted profile fields.
FR-UM-03	The system shall support system roles, project-membership roles, and independently managed permissions.
FR-UM-04	The system shall support organization-scoped user and project data.
FR-UM-05	Authorized managers shall be able to provision individual users and perform validated bulk member import.
FR-UM-06	The system shall support cohort, batch, and sub-batch attributes for member organization and filtering.

6.2 Project and Work Management

ID	Requirement
FR-WM-01	Authorized users shall create and manage projects within their permissions.
FR-WM-02	The system shall support Project → Epic → Story/Bug → Task → Subtask with enforced parent-child rules.
FR-WM-03	Authorized users shall create, assign, update, and view work items within their permitted scope.
FR-WM-04	Work items shall track title, description, type, priority, assignee(s), dates, status, progress, and story points where applicable.
FR-WM-05	The system shall identify overdue and blocked work.
FR-WM-06	The system shall provide project and work progress visibility to authorized users.
FR-WM-07	The system shall provide Roadmap, Backlog/Epics, Sprint Workspace, and Kanban views over shared work-item records.
FR-WM-08	The system shall support sprint creation, activation, completion, and unfinished-work rollover.
FR-WM-09	The system shall support project document upload and access where authorized.

6.3 Forms and Submissions

ID	Requirement
FR-FM-01	Authorized users shall create and publish forms with questions, target users or roles, and submission deadlines.
FR-FM-02	The system shall track each targeted user's submission as submitted, pending, or overdue.
FR-FM-03	Authorized users shall view targeted users, submissions, answers, and pending/overdue counts.
FR-FM-04	The system shall support reminders for pending or overdue submissions; reminders must create a real notification or clearly report that automated delivery is unavailable.

6.4 Notifications and Alerts

ID	Requirement
FR-NT-01	The system shall provide in-app notifications for relevant announcements and work/form events.
FR-NT-02	Users shall be able to view notifications and mark them as read.
FR-NT-03	The system should notify users of upcoming or missed deadlines through an approved scheduled mechanism; automated deadline scheduling is a known implementation gap.

6.5 Action Items Management (Replaces MoM)

Action Items are specific follow-up commitments agreed during a discussion and intended to be completed before the next meeting. They are not meeting minutes, a transcript, a general agenda, or a replacement for the platform's regular project tasks.

ID	Requirement
FR-AI-01	Authorized users shall record an action item describing what must be done before the next meeting.
FR-AI-02	Each action item shall have a responsible owner; additional assignees may be supported where required.
FR-AI-03	Each action item shall support a target date, status, and relevant project context.
FR-AI-04	Users shall be able to update action-item status, and authorized leads shall be able to review outstanding and completed items.
FR-AI-05	At the next meeting, users shall be able to review outstanding commitments and identify completed, pending, or overdue items.
FR-AI-06	Action items shall be persisted in the shared backend/database. A frontend-only in-memory implementation does not satisfy this requirement.

6.6 Time Tracking and Timesheets

ID	Requirement
FR-TS-01	Authorized users shall be able to start and stop a work session against an eligible work item.
FR-TS-02	The system shall record work-session start/end times and create or associate time entries.
FR-TS-03	Users shall be able to record permitted manual time entries with duration, date, task, and optional notes.
FR-TS-04	The system shall provide personal and manager timesheet summaries within authorized scope.
FR-TS-05	Timesheet calculations shall avoid double-counting overlapping work sessions and shall apply approved office-hour and lunch-break policies when calculating attendance-derived minutes.

6.7 Attendance

ID	Requirement
FR-AT-01	The system shall provide authorized visibility into work-session-derived attendance/time summaries.
FR-AT-02	The attendance policy shall define how office hours, lunch breaks, missing sessions, and non-working days affect calculated attendance.
FR-AT-03	A dedicated attendance record/check-in workflow shall only be considered in scope after the attendance model is confirmed.

6.8 Dashboards and Reporting

ID	Requirement
FR-RP-01	The system shall provide role-appropriate dashboards restricted by organization and authorized scope.
FR-RP-02	The manager view shall show total interns/members, active projects, and relevant work counts.
FR-RP-03	The system shall provide visibility into active, completed, pending, blocked, and overdue work.
FR-RP-04	The system shall provide project, roadmap, sprint, and progress visibility based on actual persisted data.
FR-RP-05	The intern view shall show assigned work, deadlines, status, forms, notifications, and available time-tracking features.
FR-RP-06	Advanced performance scoring and analytics shall not be represented as implemented until the reporting module is backed by verified calculations and data.

7. Action Items Business Rules

- An action item represents a commitment to do something before the next meeting.
- Every action item must state a concrete follow-up action and a responsible owner.

- A target date should align with the next meeting date or an explicitly agreed earlier deadline.
- Statuses should distinguish at minimum Pending, In Progress, and Completed; overdue is derived from the target date and non-completed status.
- Outstanding action items should be reviewed at the next meeting and carried forward or closed according to the team's decision.
- Action Items are not MoM. The system is not required by this BRD to store meeting transcripts, narrative meeting notes, or full meeting minutes.
- Action Items must be persisted in the shared backend and scoped to the relevant organization/project; in-memory frontend state is insufficient.

8. Dashboards and Reporting Requirements

8.1 Manager / Oversight View

- Total interns and relevant member counts, including authorized cohort/batch filtering.
- Active projects and project health/progress.
- Active, completed, pending, blocked, and overdue work items.
- Upcoming deadlines and sprint progress.
- Form publication, submission, pending, and overdue visibility.
- Timesheet and work-session summaries within authorized scope.
- Outstanding Action Items and their owners, due dates, and statuses.
- Attendance-derived time summaries, with clear indication of the calculation policy.
- Do not present advanced performance reports as available while the reports screen remains a placeholder.

8.2 Intern / Employee View

- Assigned work items and permitted project context.
- Priority, deadlines, status, and hierarchy context.
- Assigned forms, submission state, and notifications.
- Start/stop work tracking and personal time entries where enabled.
- Action Items assigned to the user, including due date and status.
- Profile and authorized project/sprint visibility.

8.3 Project Workspaces

Roadmap, Backlog/Epics, Sprint Workspace, and Kanban shall use the same canonical work-item and sprint data. Changes made in one view should be reflected in other views that show the same records.

9. Business Rules

- Organization boundaries apply to all organization-owned projects, work items, sprints, forms, notifications, documents, and time entries.
- Work items shall follow the hierarchy and leaf rules in Section 3.
- Manager personal tasks shall remain separate from project work-item hierarchies.
- An item is overdue when its applicable due date has passed and it is not completed; equivalent rules apply to form submissions and Action Items.
- Only one sprint may be active per project at a time.
- Unfinished sprint items shall be moved to the backlog or an eligible target sprint when an active sprint is completed.
- Timesheet duration and attendance-derived minutes shall follow approved overlap, office-hours, timezone, and lunch-break rules.
- Notifications and reminders must not be represented as delivered when only a UI toast was shown.
- All sensitive actions must be authorized server-side.

10. Non-Functional Requirements

ID	Requirement
NFR-01 Security	Authentication, password protection, role/permission checks, and organization-level data isolation shall be enforced by the backend.
NFR-02 Data integrity	Relational constraints and transactional operations shall preserve project, hierarchy, sprint, and time-entry consistency.
NFR-03 Consistency	All project views shall read from the shared source of truth; optimistic UI updates must recover correctly when API requests fail.
NFR-04 Session handling	The design proposes a 15-minute inactivity timeout. Implementation audit reports the client-side sliding inactivity behavior is not fully wired; this requirement remains unverified until tested end-to-end.
NFR-05 Performance	The HLD states a target of under 150 ms for API queries, but no benchmark evidence was provided in the audit. Treat this as a target requiring measurement, not a verified result.
NFR-06 Resilience	If the notification queue/cache is unavailable, the application shall fail safely and preserve notification data where the configured fallback supports it.
NFR-07 Usability	Directory and work views shall support search/filtering and usable pagination or grouping as member/work volume grows.
NFR-08 Auditability	Work assignments, status, sprint rollover, time entries, form submissions, and Action Item status must be traceable through persisted records where supported.

11. Implementation Status and BRD Traceability

This section is a traceability snapshot based on the implementation audit supplied on October 9, 2026. It is not a substitute for a fresh runtime test. 'Implemented' reflects the audit's findings; 'Partial' and 'UI only' must not be presented as complete functionality.

Capability	Audit status	BRD treatment / gap
Authentication and role-based routing	Implemented	Retain; verify role mapping and access matrix.
Member directory, search, provisioning, bulk import	Implemented	Include in scope; confirm UI pagination behavior separately.
Cohort management/filtering	Implemented	Include in requirements and preserve organization boundaries.
Projects and project documents	Implemented	Include.
Hierarchy, backlog, Kanban, roadmap, sprint lifecycle	Implemented	Include strict hierarchy, BUG support, story points, and rollover.
Sprint burndown/health	Implemented in audit, but metrics require correctness checks	Specify calculations; validate empty/one-task/reopened/future-date cases.
Forms engine and submissions	Implemented	Include; pending/overdue reminder action is UI-only.
In-app notifications	Implemented/partial triggers	Include in-app delivery; event-trigger and deadline automation gaps remain.
Timesheets and work sessions	Implemented	Include work-session and time-entry requirements.
Attendance standalone UI	UI only / partial derived metrics	Do not claim a complete attendance workflow.
Manager performance reports	UI only	Not an implemented reporting capability.
Action Items UI	Partially implemented; in-memory	Requirement is defined, but backend persistence remains a gap.
Intern profile editing	Partially implemented	Only verified basic fields; avatar upload not implemented.
Sliding inactivity timeout	Unverified / incomplete	HLD claims it; audit says client behavior is not fully wired.
Hierarchy tests	30/30 passed per audit	Retain the verified test result with date and scope.
Timesheet verification	Passed per audit	Retain the verified test result with date and scope.
Frontend build and backend typecheck	Passed per audit	Retain as point-in-time results, not a guarantee of future builds.

11.1 Cross-Document Inconsistencies to Resolve

- HLD lists Tailwind CSS 4 while the implementation audit reports Tailwind CSS 3.4; verify package.json and align the HLD.
- HLD states PostgreSQL 15 while the audit reports PostgreSQL 16 / Supabase; verify the actual deployment/runtime version and document it accurately.
- HLD says sliding 15-minute inactivity timeout is implemented; audit says the client-side inactivity listener is not fully wired. Re-test and correct the HLD.

- HLD claims there are no client-only placeholders, but the audit identifies placeholder Reports and Attendance screens and in-memory Action Items/attendance behavior. Correct the HLD to distinguish implemented from partial modules.
- HLD describes Action Items in terms of sync notes and meeting decisions; revise it to focus on follow-up commitments due before the next meeting, not MoM.
- LLD's SprintStatus enum lists PLANNED, ACTIVE, COMPLETED, while the audit reports CANCELLED as an additional status. Verify the Prisma enum and align the LLD.
- LLD's verification matrix marks descendant deletion and several sprint/personal-task edge cases as Planned. Do not describe these tests as passed until implemented and run.
- LLD says hierarchy tests passed before its revision while also listing additional planned tests. Clearly separate completed test cases from planned coverage.
- LLD/HLD role descriptions do not consistently match the audit's six system roles plus project-level roles. Publish one authoritative role and permission matrix.
- BRD v2.1 lists Technical Architect, Capability Lead and Mentor as roles, but the audit did not find them as implemented system roles. Obtain a business decision before treating them as implemented.

12. Open Decisions for Mentor / Product Owner

1. Confirm the mapping from business titles (Technical Architect, Capability Lead, Mentor) to implemented system/project roles, or approve new roles.
2. Choose the attendance model: retain policy-derived attendance from work sessions, or require a dedicated attendance record/check-in workflow. Specify exceptions and manager overrides if needed.
3. Approve Action Items as a distinct follow-up feature: owner, target date, status, project context, and review at the next meeting. Confirm whether a next-meeting date/reference must be stored.
4. Approve when and how action items should notify owners; automatic reminders are not verified as operational.
5. Define final work-item status transitions and project/sprint status values, including whether CANCELLED is a supported sprint status.
6. Confirm the progress formula for projects and Epics, and the burndown unit when some items lack story-point estimates.
7. Define the requirements and calculation rules for any future performance scoring and analytics.
8. Confirm whether manual attendance, non-working days, leave, and missing work sessions must be represented.
9. Confirm the inactivity timeout policy and require an end-to-end test before claiming it is implemented.

13. Summary

BRD v3.0 expands the original v2.1 business scope to reflect the current platform's organization-scoped user management, cohort filtering, Jira-style hierarchy with Bug support, synchronized project workspaces, sprint lifecycle, work-session tracking, timesheets, forms, notifications, and Action Items. It explicitly distinguishes implemented functionality from partial or UI-only modules. The BRD defines Action Items as follow-up commitments to complete before the next meeting and excludes MoM authoring. Open role, attendance, notification, and progress decisions remain subject to mentor confirmation.